MULTIPLE INTEGRALS & IT'S APPLICATIONS

Sec – A (1 mark)

Q1. The value of the integral $\int_0^1 \int_0^3 (x+5)dydx$ **is:**

A. $\frac{13}{2}$

B. $\frac{33}{2}$

C. $\frac{33}{4}$

D. $\frac{23}{2}$

Q2. The value of integral $\int_0^1 \int_0^3 (3x^2 + 3y^2)dx\,dy$ **is**

A. 30

B. 28

C. 32

D. 27

Q3. The value of integral $\int_0^{\pi/2} \int_0^a r\,dr\,d\theta$ **is:**

A. $\frac{\pi a^2}{8}$

B. $\frac{\pi a^2}{4}$

C. $\frac{\pi a^2}{2}$

D. None of these

Q4. The value of the integral $\int_0^1 \int_0^x \int_0^{x-y} dz\,dy\,dx$ **is:**

A. $\frac{1}{2}$

B. $\frac{1}{3}$

C. $\frac{1}{4}$

D. $\frac{1}{6}$

Q5. The value of the integral $\int_0^a \int_0^x \int_0^{x-y} x\,dx\,dy\,dz$ is:

A. $\frac{a^4}{6}$

B. $\frac{a^4}{8}$

C. $\frac{a^4}{4}$

D. $\frac{a^4}{2}$

Q6. When we change from Cartesian to spherical polar coordinates, the Jacobian of the transformation is equal to

A. r

B. $r\sin\theta$

C. $r\cos\theta$

D. $r^2\sin\theta$

Q7. For *x>0*, the Gamma function $\Gamma(x+1)$ is:

A. $x\Gamma x$

B. $(x+1)\Gamma x$

C. $x\Gamma(x+1)$

D. $x(x+1)$

Q8. The Gamma function, $\Gamma x = \int_0^\infty e^{-t}t^{x-1}dt$ is valid only for:

A. x>0

B. x = 0

C. x<0

D. for all x

Q9. The Beta function defined by $B(x,y) = \int_0^1 t^{x-1}(1-t)^{y-1}dt$ is valid for

A. x>0 and y<0

B. x<0 and y>0

C. x<0 and y<0

D. x>0 and y>0

Q10. Which of the following statement is not correct?

A. B(x, y) = B(y, x)

B. $\Gamma 1 = 1$

C. B(x, y) = $\Gamma x\, \Gamma y$

D. $\Gamma x = (x-1)!$, if x is a non-negative integer

Q11. The value of the integral $\iint_R xy\, dx\, dy$ over the quarter circle $R = \{(x,y): 0 \leq x, 0 \leq y, x^2 + y^2 \leq 1\}$ is :

A. $\frac{1}{4}$

B. $\frac{1}{8}$

C. $\frac{1}{2}$

D. $\frac{1}{3}$

Q12. The value of the integral $\int_0^1 \int_0^{\sqrt{1+x^2}} \frac{dydx}{1+x^2+y^2}$ is:

A. $\frac{\pi}{4}\log\left(\sqrt{2}+1\right)$

B. $\frac{\pi}{4}\log\left(\sqrt{2}-1\right)$

C. $\frac{\pi}{2}\log\left(\sqrt{2}+1\right)$

D. $\frac{\pi}{2}\log\left(\sqrt{2}-1\right)$

Q13. The value of the integral $\int_0^{\pi/2}\int_0^{\pi/2}\sin(x+y)dx\,dy$ is

A. 0

B. π

C. $\frac{\pi}{2}$

D. 2

Q14. A triangle ABC consists of vertex points A(0,0), B(1,0) and C(0,1). The value of integral $\iint 2x\,dx\,dy$ over the triangle is :

A. 1

B. $\frac{1}{3}$

C. $\frac{1}{8}$

D. $\frac{1}{9}$

Q15. The value of the integral $\int_0^1\int_0^{1-x}\int_0^{2-x} xyzdzdydx$ is:

A. $\frac{11}{240}$

B. $\frac{13}{120}$

C. $\frac{13}{240}$

D. $\frac{11}{120}$

Q16. The value of the integral $\int_0^3\int_0^{3-x}\int_0^{3-x-y} dx\,dy\,dz$ is:

A. $\frac{9}{2}$

B. $\frac{9}{4}$

C. $\frac{3}{2}$

D. $\frac{7}{2}$

Q17. By the change of order of integration the integral $\int_0^{4a}\int_{x^2/4a}^{2\sqrt{ax}} dy\,dx$ leads to :

A. $\int_0^{4a}\int_{y^2/4a}^{ay} dx\,dy$

B. $\int_0^{4a}\int_0^{2\sqrt{ay}} dx\,dy$

C. $\int_0^{4a}\int_{y^2/4a}^{2\sqrt{ay}} dx\,dy$

D. $\int_0^{a}\int_{y^2/4a}^{\sqrt{ay}} dx\,dy$

Q18. The value of B(n, n+1) is:

A. $\frac{\Gamma(n)^2}{n\Gamma n}$

B. $\frac{\Gamma(n+1)}{\Gamma n\Gamma 2n}$

C. $\frac{\Gamma n}{\Gamma n\Gamma(n+1)}$

D. $\frac{\Gamma(n)^2}{2\Gamma(2n)}$

Q19. The value of the integral $\int_0^{\pi/2}\sqrt{tan\,\theta}\,d\theta$ is:

A. $\frac{\pi}{2}$

B. $-\frac{\pi}{\sqrt{2}}$

C. π

D. $\frac{\pi}{\sqrt{2}}$

Q20. If $B(n,2)=\frac{1}{6}$ and n is a positive integer then value of n is

A. 3

B. -2

C. 2

D. -3

Q21. The value of the integral $\int_0^1 \int_0^{\sqrt{\frac{1-y^2}{2}}} \frac{dxdy}{\sqrt{1-x^2-y^2}}$ is:

A. $\frac{\pi}{4}$

B. $\frac{\pi}{2}$

C. $-\frac{\pi}{4}$

D. $-\frac{\pi}{2}$

Q22. The value of the integral $\int_0^u \int_0^{u-x} \int_0^{u-x-y} dx\, dy\, dz$ is:

A. $\frac{a^2}{2}$

B. $\frac{a^2}{4}$

C. $-\frac{a^2}{2}$

D. $-\frac{a^2}{4}$

Q23. The value of the integral $\iint (x^2 - y^2) dx\, dy$ over the triangle with vertices (0,1) , (1,1) and (1,2)

A. $\frac{-2}{3}$

B. $\frac{2}{3}$

C. $\frac{4}{3}$

D. $-\frac{4}{3}$

Q24. The value of B(x, y+1) is equal to:

A. $\frac{x}{x+y} B(x, y)$

B. $\frac{y}{x+y}\ B(x,y)$

C. $\frac{x+y}{x}\ B(x,y)$

D. $\frac{x+y}{y}\ B(x,y)$

Q25. Which of the following statement is incorrect?

A. In double integration, if the limits of integration are constants, then the order of integration is immaterial; provided the limits of integration are changed accordingly.

B. In double integration with variable limits, the change of order of integration does not affect the limits of integration.

C. The change of order of integration does not in any way affect the value of integral.

D. In double integration with variable limits, the change of order of integration affect the limits of integration.

Sec – B (2 marks)

Q1. By double integration, the area enclosed by the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is :

A. πab

B. $\frac{\pi ab}{2}$

C. $\frac{\pi ab}{4}$

D. $2\pi ab$

Q2. The value of the integral $\int_0^a \int_0^{\sqrt{a^2-x^2}} \int_0^{\sqrt{a^2-x^2-y^2}} dx\,dy\,dz$ is:

A. $\frac{\pi a^3}{2}$

B. $\frac{\pi a^3}{4}$

C. $\frac{2\pi a^3}{3}$

D. $\frac{\pi a^3}{6}$

Q3. The value of the integral $\int_0^1 \int_y^1 e^{-x^2} dx\, dy$ is:

A. 1

B. -1

C. $\frac{1}{2}$

D. 2

Q4. The value of $\iint xy(x+y)dx\, dy$ over the area between $y = x^2$ and $y = x$ is

A. $\frac{3}{41}$

B. $\frac{3}{56}$

C. $\frac{2}{56}$

D. $\frac{2}{41}$

Q5. The smaller of the areas bounded by the circle $x^2 + y^2 = 9$ and the line $x + y = 3$ is

A. $\frac{9}{2}(\pi - 2)$ Sq. units

B. $\frac{9}{4}(\pi + 2)$ Sq. units

C. $\frac{9}{4}(\pi - 2)$ Sq. units

D. $\frac{9}{2}(\pi + 2)$ Sq. units

Q6. The volume of the tetrahedron bounded by the coordinate planes and the plane $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$

is:

A. abc

B. $\frac{abc}{2}$

C. $\frac{abc}{3}$

D. $\frac{abc}{6}$

Q7. The value of the integral $\int_0^{\pi/2} sin^7\theta\, cos^9\theta\, d\theta$ is:

A. $\frac{1}{280}$

B. $\frac{1}{560}$

C. $\frac{1}{140}$

D. $\frac{1}{640}$

Q8. The value of the integral $\int_0^{\frac{\pi}{2}} \int_0^{a\sin\theta} \int_0^{a^2-r^2} r dz dr d\theta$ is:

A. $\frac{5\pi a^4}{64}$

B. $\frac{\pi a^4}{64}$

C. $\frac{5\pi a^4}{32}$

D. $\frac{3\pi a^4}{64}$

Q9. The area bounded by the parabola $y = x^2$ and the line $y = 2x + 3$ is:

A. $\frac{10}{3}$ Sq. units

B. $\frac{20}{3}$ Sq. units

C. $\frac{8}{3}$ Sq. units

D. $\frac{16}{3}$ Sq. units

Q10. The value of integral $\int_0^{\infty} \int_x^{\infty} \frac{e^{-y}}{y} dy\, dx$ is equal to :

A. 0

B. 2

C. -1

D. 1

Q11. By changing the order of integration, the integral $\int_{-a}^{a}\int_{0}^{\sqrt{a^2-x^2}} f(x,y)dxdy$ leads to :

A. $\int_{-a}^{a}\int_{0}^{\sqrt{a^2-x^2}} f(x,y)dydx$

B. $\int_{0}^{a}\int_{0}^{\sqrt{a^2-x^2}} f(x,y)dydx$

C. $\int_{0}^{a}\int_{-\sqrt{a^2-x^2}}^{\sqrt{a^2-x^2}} f(x,y)dydx$

D. $\int_{-a}^{a}\int_{-\sqrt{a^2-x^2}}^{\sqrt{a^2-x^2}} f(x,y)dydx$

Q12. The value of the integral $\iint \sin \pi\left(x^2+y^2\right)dxdy$ over the circle $x^2+y^2 \leq 1$ is equal to:

A. π

B. 2

C. 1

D. 4

Q13. The value of the integral $\iint (a-x)^2 dx\, dy$ over the right half of the circle $x^2+y^2=a^2$

A. $a^4\left(\frac{5\pi}{8}-\frac{4}{3}\right)$

B. $a^4\left(\frac{5\pi}{8}+\frac{4}{3}\right)$

C. $a^4\left(-\frac{5\pi}{8}+\frac{4}{3}\right)$

D. $a^4\left(-\frac{5\pi}{8}-\frac{4}{3}\right)$

Q14. The value of the integral $\int_0^{\pi/2}\sqrt{\cot\theta}\, d\theta$ is :

A. $\frac{\pi\sqrt{2}}{2}$

B. $\frac{\pi}{2}$

C. $\frac{\pi\sqrt{2}}{4}$

D. $\frac{\pi}{4}$

Q15. The value of the integral $\int_0^\infty t^{3/2}(1-e^{-t})dt$ is:

A. 2π

B. $\sqrt{\pi}$

C. $\sqrt{2\pi}$

D. $2\sqrt{\pi}$

Q16. The value of $\iint_R ye^{-xy}\,dx\,dy$ where $R=[0,2]\times[0,3]$ is:

A. $\frac{5}{2}-\frac{1}{2}e^{-6}$

B. $\frac{5}{2}+\frac{1}{2}e^{-6}$

C. $\frac{3}{2}+\frac{1}{2}e^{-6}$

D. $\frac{3}{2}-\frac{1}{2}e^{-6}$

Q17. The area outside the circle $r=a$ and inside the cardioids $r=a(1+\cos\theta)$ is :

A. $\frac{a^2}{2}(\pi+8)$ Sq. units

B. $\frac{a^2}{2}(\pi-8)$ Sq. units

C. $\frac{a^2}{4}(\pi+8)$ Sq. units

D. $\frac{a^2}{4}(\pi-8)$ Sq. units

Q18. The value of the integral $\iint r^2 dr d\theta$ over the area included between the circles $r=2\sin\theta$ and $r=4\sin\theta$ is:

A. $\frac{45\pi}{2}$

B. $\frac{45\pi}{4}$

C. 45π

D. $\frac{15\pi}{3}$

Q19. The value of the integral $\int_{-1}^{1}\int_{0}^{z}\int_{x-z}^{x+z}(x+y+z)dydxdz$ **is:**

A. 3

B. 2

C. 1

D. 0

Q20. The value of the integral $\int_0^1 \frac{1}{\sqrt[3]{1-x^3}}dx$ **is:**

A. $\frac{2\pi}{3}$

B. $\frac{2\pi}{\sqrt{3}}$

C. $\frac{\pi}{3\sqrt{3}}$

D. $\frac{2\pi}{3\sqrt{3}}$

Q21. The value of the integral $\int_0^{\pi/2} sin^{11}\theta\ cos^3\theta\ d\theta$ **is:**

A. $\frac{1}{84}$

B. $\frac{1}{42}$

C. $\frac{1}{168}$

D. $\frac{1}{82}$

Q22. The value of integral $\int_0^1 \int_{e^x}^{e} \frac{1}{\log y} dy\, dx$ is equal to

A. $e-2$

B. $e-1$

C. $2e-1$

D. $e+1$

Q23. The value of the integral $\iint r^2 dr d\theta$ over the area included between the circles $r = 2\sin\theta$ and $r = 4\sin\theta$ is:

A. $\frac{45\pi}{2}$

B. $\frac{45\pi}{4}$

C. 45π

D. $\frac{15\pi}{3}$

Q24. The area lying between the parabola $y = 4x - x^2$ and the line $y = x$ is

A. 3 sq. units

B. 4 sq. units

C. 3.5 sq. units

D. 4.5 sq. Units

Q25. The value of the integral $\iint_R y dx dy$ where *R* is the region bounded by the parabolas $y^2 = 4x$ and $x^2 = 4y$.

A. $\frac{48}{5}$

B. $\frac{24}{5}$

C. $\frac{42}{5}$

D. $\frac{54}{5}$